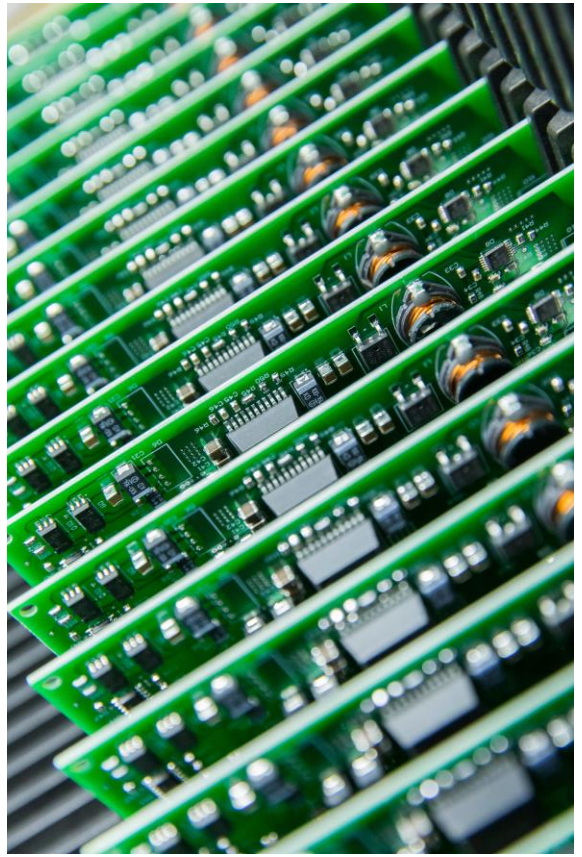
Natural Convection Between Parallel Plates

Natural Convection – Lesson 4

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Natural Convection in Parallel Plate Channels

- Understanding natural convection in parallel plate channels is important for many real-world applications, for example, when analyzing heat transfer from multiple printed circuit boards stacked parallel to each other or heat exchange in radiators using large parallel fins.
- Such parallel plate channels are categorized based on their orientation as:
  - Vertical: Buoyancy force acts only in the vertical direction and is responsible for establishing the fluid flow in this direction.
  - Inclined: The buoyancy force has two components (normal and streamwise) which are responsible for strong secondary flows.

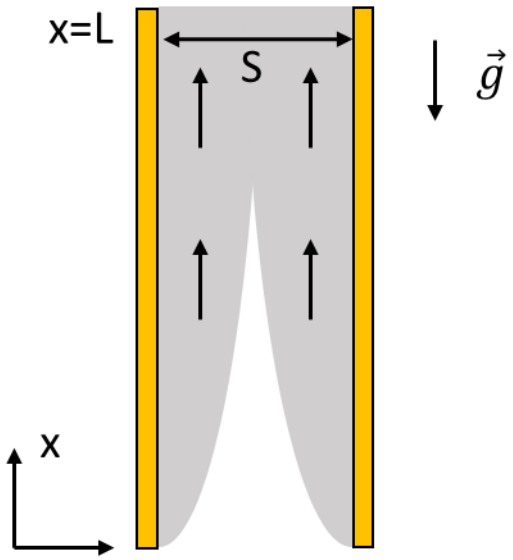


Multiple PCBs inside electronics forming parallel plate channels

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Natural Convection in Parallel Plate Channels (cont.)

- For short channels and/or channels with large spacing (small L/S)
  - Boundary layers develop independently over each plate.
  - Heat transfer rates are estimated based on external flow relationships developed for isolated plates in quiescent fluid.
- For long channels and/or channels with narrow spacing (large L/S)
  - Boundary layers on each plate eventually merge into a fully developed flow.
  - New relationships are necessary to predict the heat transfer rate.
- Surface conditions for such flows are extremely critical. Two conditions are discussed in this lesson:
  - Isothermal or Isoflux and Symmetrical ($T_{s,1} = T_{s,2}; q''_{s,1} = q''_{s,2}$)
  - Isothermal or Isoflux and Asymmetrical($T_{s,1} \neq T_{s,2}; q''_{s,1} \neq q''_{s,2}$)



Vertical Parallel Plates at Large L/S

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Vertical Channel – Isothermal Conditions

- Elenbaas' correlation for symmetrically heated isothermal ($T_{s,1}$ and $T_{s,2}$) plates proposes the following relationship for the average Nusselt number.

$$\overline{Nu}_S = \frac{1}{24} Ra_S \left(\frac{S}{L}\right)\left(1 - e^{\left(-\frac{35}{Ra_S(S/L)}\right)}\right)^{3/4}$$

For $10^{-1} \lesssim \frac{S}{L} Ra_S \lesssim 10^5$

Applicable for air

Where S is the space between the plates and L is the length of the channel. The Rayleigh number is estimated using S as the characteristic length:

$$Ra_S = \frac{g\beta(T_s - T_\infty)S^3}{\alpha\nu}$$

- The average Nusselt number for isothermal conditions can be defined as:

$$\overline{Nu}_S = \frac{qS}{A(T_s - T_\infty)k}$$

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Vertical Channel – Isothermal Conditions (cont.)

- As $S/L \to 0$ the fully developed limit is reached. At this condition, the correlation for the average Nusselt number reduces to:

$$\overline{Nu}_{S(fd)} = \frac{Ra_S}{24}\left(\frac{S}{L}\right)$$

The subscript "fd" indicates that flow is fully developed.

- A similar fully developed limit can be obtained for an asymmetric heating condition – plate 1: Isothermal $(T_{s,1})$ and plate 2: Adiabatic $(q"_{s,2} = 0)$. The average Nusselt number for the isothermal plate is:

$$\overline{Nu}_{S(fd)} = \frac{Ra_S}{12}\left(\frac{S}{L}\right)$$

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Vertical Channel – Isothermal Conditions (cont.)

- Using the correlations for the fully-developed limit and the isothermal plate conditions, Bar-Cohen and Rohsenow obtained the following correlation which is applicable for a complete range of S/L.

Fluid properties are estimated at $\bar{T} = (T_s + T_\infty)/2$

$$\overline{Nu}_S = \left\{ \left(\frac{C_1}{\left(Ra_S \frac{S}{L} \right)^2} \right) + \left(\frac{C_2}{\left(Ra_S \frac{S}{L} \right)^{1/2}} \right) \right\}^{-1/2}$$

Fully developed: $Ra_S \frac{S}{L} \lesssim 10$

Isolated Plate limit: $Ra_S \frac{S}{L} \gtrsim 100$

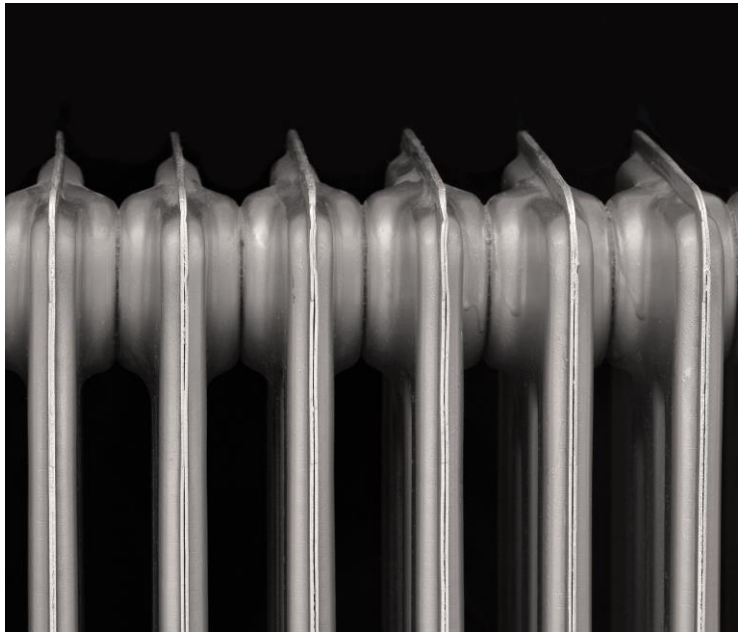
- The table below reports the values of the coefficients C_1 and C_2 that can be used for different isothermal surface conditions:

Surface Condition	C_1	C_2
Symmetric Isothermal plates ($T_{s,1} = T_{s,2}$)	576	2.87
Asymmetric Isothermal/Adiabatic plates ($T_{s,1}; q''_{s,2} = 0$)	144	2.87

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Array of Vertical Channels – Isothermal Conditions

- The fluid flow driven by the buoyancy effects between the vertical plates must overcome the viscous forces imposed by the walls. The fluid mass flow rate is governed by the balance between these buoyancy and viscous forces.
- The viscous force increases with decreasing plate spacing (S), and this decreases the overall mass flow rate between the plates.
- A decrease in mass flow rate reduces the average heat transfer coefficient. From the previous relationship, the average Nusselt number from each plate decreases with decreasing value of spacing between the plates (S).
- An optimal plate spacing (S_{opt}) to maximize heat transfer for an array of isothermal vertical plates is proposed using the previous relationship.
- For an array of vertical parallel plates (say, fins) in a fixed width W, the number of plates increases with decreasing spacing between the plates (S).
- This increases the total plate surface area of heat transfer, A_s, (for a prescribed volume). The product of $\bar{h}$ and A_s is maximum in an array at $S = S_{opt}$, maximizing the heat transfer.
  - For $S < S_{opt}$, the reduction of $\bar{h}$ due to viscous effects overcomes the increase of A_s.
  - For $S > S_{opt}$, the decrease in the total number of plates reduces A_s and overcomes the benefit of an increased $\bar{h}$.



Fins of a radiator forming parallel plate channels

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Array of Vertical Channels – Isothermal Conditions (cont.)

- The heat transfer from each individual plate is maximized at $S = S_{max}$.
- To obtain this maximum heat transfer from each plate, S_{max} must be large enough to prevent the merging of boundary layers. This means that the heat transfer solution should tend towards the isolated plate limit.
- The table below reports the relations to estimate the optimal spacing for isothermal conditions.

Surface Condition	S_{opt}	$\frac{S_{max}}{S_{opt}}$
Symmetric Isothermal plates ($T_{s,1} = T_{s,2}$)	$2.71\left(\frac{Ra_S}{S^3 L}\right)^{-1/4}$	1.71
Asymmetric Isothermal/Adiabatic plates ($T_{s,1}; q''_{s,2} = 0$)	$2.15\left(\frac{Ra_s}{S^3 L}\right)^{-1/4}$	1.71

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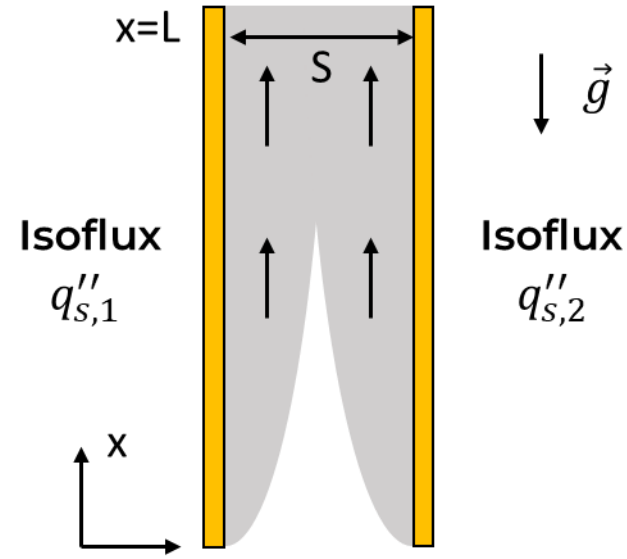
Vertical Channel – Constant Heat Flux Condition

- In the constant heat flux case, we change the definition of the local Nusselt and Rayleigh numbers to the following:

$$Nu_{S,L} = \frac{q_s'' S}{(T_{s,L} - T_\infty)k}$$

$T_{s,L}$ refers to the plate temperature at $x = L$

$$Ra_s^* = \frac{g\beta q_s'' S^4}{k\alpha\nu}$$



Vertical Parallel Plates at Isoflux Conditions

- Under symmetric conditions ($q''_{s,1} = q''_{s,2}$), the fully developed limit for the Nusselt number is:

$$Nu_{S,L(fd)} = 0.144\left[Ra_s^*\left(\frac{S}{L}\right)\right]^{1/2}$$

- A similar limit is obtained for an asymmetric heating condition – plate 1: Constant heat flux $\left(q''_{s,1}\right)$ and plate 2: Adiabatic ($q''_{s,2} = 0$)

$$Nu_{S,L(fd)} = 0.204\left[Ra_s^*\left(\frac{S}{L}\right)\right]^{1/2}$$

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Vertical Channel – Constant Heat Flux Condition (cont.)

- A similar correlation for the local Nusselt number for the complete range of S/L is obtained by Bar-Cohen and Rohsenow.

Fluid properties are estimated at $\bar{T} = (T_{s,\mathrm{L}} + T_\infty)/2$

$$Nu_{S,L} = \left\{ \left(\frac{C_1}{\left(Ra_s^* \frac{S}{L} \right)} \right) + \left(\frac{C_2}{\left(Ra_s^* \frac{S}{L} \right)^{2/5}} \right) \right\}^{-1/2}$$

Fully developed: $Ra_s^* \frac{S}{L} \lesssim 10$

Isolated Plate limit: $Ra_s^* \frac{S}{L} \gtrsim 100$

- For an array of vertical plates with constant heat flux condition, the total volumetric heating rate increases with decreasing plate spacing.
- The primary objective of the heat transfer analysis with constant wall flux condition is to maintain the surface temperature value below a certain prescribed limit. Therefore, the optimal plate spacing (S_{opt}) is defined based on maximum volumetric heat dissipation per unit temperature difference $(T_s(L) - T_\infty)$.
- For a prescribed heat flux value, the maximum spacing (S_{max}) is defined as the spacing between the plates that leads to the lowest possible surface temperature (without accounting for any volumetric considerations).

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Vertical Channel – Constant Heat Flux Condition (cont.)

- The table below reports the values of the coefficients C_1 and C_2, and the relation to estimate the optimal spacing for different isothermal surface conditions:

Surface Condition	C_1	C_2	S_{opt}	$\frac{S_{max}}{S_{opt}}$
Symmetric plates with Constant Heat Flux ($q''_{s,1} = q''_{s,2}$)	48	2.51	$2.12\left(\frac{Ra_S^*}{S^4 L}\right)^{-1/5}$	4.77
Asymmetric Isoflux/Adiabatic plates ($q''_{s,1}; q''_{s,2} = 0$)	24	2.51	$1.69\left(\frac{Ra_S^*}{S^4 L}\right)^{-1/5}$	4.77

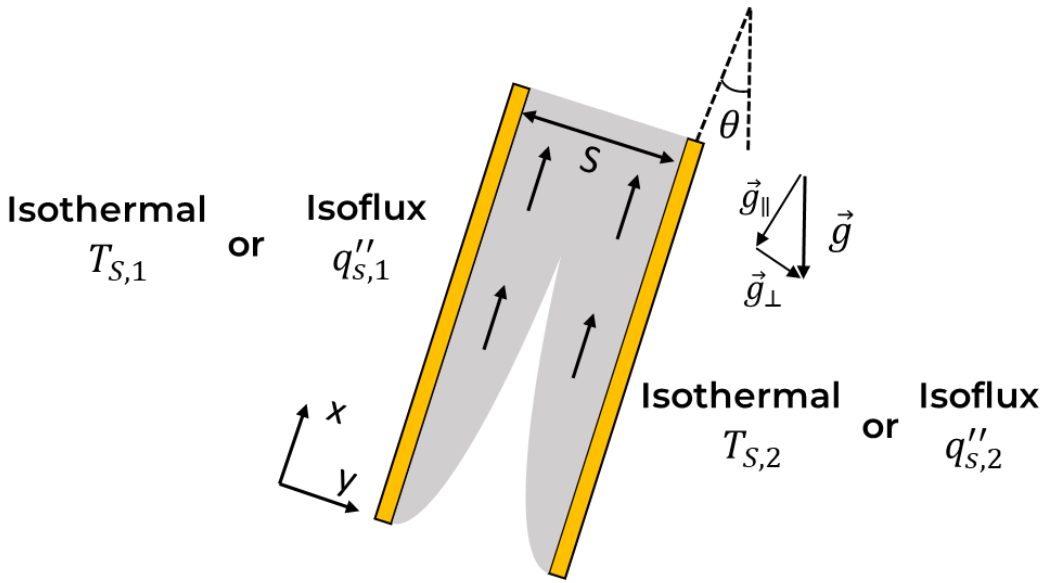
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Inclined Channels

- The heat transfer relationship for inclined channels was obtained by experiments conducted by Sparrow and Azevedo using water as the fluid.
- Small angles of inclination $0° \leq \theta \leq 45°$ were tested under both symmetric isothermal ($T_{s,1} = T_{s,2}$) and asymmetric isothermal – adiabatic ($T_{s,1}; q"_{s,2} = 0$) conditions.
- The following correlation for the average Nusselt number was obtained.

$$\overline{Nu}_S = 0.645 \left[Ra_S \left(\frac{S}{L} \right) \right]^{1/4}$$

Fluid properties are estimated at $\overline{T} = (T_s + T_\infty)/2$



- Strong 3D secondary flows are expected, especially for large tilt angles when the bottom surface is heated. These secondary flows tend to enhance the overall heat transfer in these inclined channels.

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Summary

- We learned about natural convection in vertical channels under symmetric and asymmetric isothermal, and constant heat flux conditions.
- We outlined the relationships for fully developed and isolated plate limits for heat transfer in each case.
- We also extrapolated this discussion to understand the effect of these conditions on an array of vertical parallel plates.
  - Isothermal condition:
    - The fluid mass flow rate reduces with decreasing plate spacing. This reduces the heat transfer coefficient.
    - Decreasing the plate spacing (under fixed volumetric considerations) increases the total surface area available for heat transfer by increasing the number of plates.
    - This gives rise to an optimal plate spacing (S_{opt}).
  - Constant Heat Flux condition:
    - The definition of optimal spacing (S_{opt}) was changed to ensure maximum heat transfer per unit change in temperature difference $(T_s(L) - T_\infty)$.

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